Combinatorial enumeration of coupled-cluster and extended coupled-cluster diagrams

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Abstract

Connected antisymmetrized diagram counts provide consistency checks for automatically generated coupled-cluster equations and operators. We restate the representation and construction algorithm of Kállay and Surján and derive equation counts from a multiset generating function. Five contraction polynomials suffice for a one- and two-body Hamiltonian. Their complete-equation sum has denominator $(1-x)^4(1+x)^2$, explaining the cubic growth and parity dependence of the counts previously obtained by Musiał, Kucharski, and Bartlett. Low-rank and finite-truncation corrections follow directly. Allowing external annihilation lines gives a quartic polynomial for the full similarity-transformed Hamiltonian; exact enumeration validates both results through cluster rank 20. We then extend the construction to a second similarity transformation by a pure deexcitation operator. Particle–hole incidence rows record each deexcitation vertex's connections to the Hamiltonian and to individual excitation vertices. Unique representatives require quotienting both row permutations and residual permutations of equivalent excitation vertices. The latter group has order at most four for the present Hamiltonian. This yields an explicit finite construction algorithm and a Burnside-averaged generating function for independent counting checks. The two successive connectivity conditions for the literal double similarity transform are distinguished from the stronger doubly-linked restriction on the energy in canonical extended coupled-cluster amplitude coordinates. Small-truncation enumerations validate the incidence construction and illustrate its increased combinatorial complexity. All counts concern generic, unfactorized antisymmetrized topologies, not labelled Wick contractions or algebraic weights.

I. Introduction

Automatic generation of coupled-cluster (CC) equations separates the combinatorial task of identifying connected contractions from their algebraic interpretation and numerical implementation. The compact representation of Kállay and Surján associates each antisymmetrized diagram with a Hamiltonian vertex and a canonical collection of cluster-vertex descriptors.[1] An independently

obtained diagram count is a useful check on this construction: it can detect missing or duplicated topologies without requiring a comparison of their algebraic expressions.

The projected-equation counting problem has already been solved for the electronic one- and two-body Hamiltonian. Musiał, Kucharski, and Bartlett derived cubic, parity-dependent formulas for both antisymmetrized and Goldstone diagrams and distinguished a fully retained equation at a specified CC truncation from a complete equation that no longer changes on increasing that truncation.[2] Their Table 1 gives the counts for three successive truncations relative to the equation rank. The present work does not propose a new count in place of those results. Its purpose is to derive the antisymmetrized count from a single multiset generating function and to make its connection to the canonical representation explicit.

The relevant mathematical tools are the ordinary generating functions for unlabelled multisets,[3] rational-function coefficient extraction,[4] and the characterization of quasipolynomials by poles at roots of unity.[5] These tools explain why the count is cubic, why even and odd equation ranks differ, and why a finite number of initial values require separate corrections. They also permit a direct treatment of the correlation-energy equation and of the bare Hamiltonian terms that are excluded from some published diagram counts. We further distinguish the projected equations from the full similarity-transformed operator: the latter retains terms that annihilate the reference but contribute when acting on excited states. Counting these terms requires relaxing the contraction constraints, not summing only the CC equations retained at a given truncation.

In extended coupled cluster (ECC), an additional exponential of a pure deexcitation operator introduces contractions between distinct cluster families. The resulting topology is no longer determined by a multiset of one-vertex contraction totals. We retain the original Hamiltonian–excitation subgraph and add an incidence matrix for the deexcitation vertices. This extension also requires distinguishing ordinary graph canonicalization from the canonical **amplitude coordinates** of ECC, whose energy obeys a stronger doubly-linked restriction.[6]

II. Counting conventions and canonical diagrams

A. Hamiltonian and equation ranks

Let the Hamiltonian, normal ordered with respect to a single-determinant reference Φ_0, be written as

$$H = E_{\mathrm{ref}} + H_{\mathrm{N}}, \quad H_{\mathrm{N}} = F_{\mathrm{N}} + W_{\mathrm{N}}, \quad T = \sum_{e=1}^{n} T_e. \tag{1}$$

Here F_{N} and W_{N} are the normal-ordered one- and two-body operators, and n is the maximum cluster excitation rank. The equation rank is k, with $0 \leq k \leq n$. We count the connected antisymmetrized topologies contributing to

$$R_k = \left\langle \Phi_k \mid (H_{\mathrm{N}} \exp T)_{\mathrm{c}} \mid \Phi_0 \right\rangle. \tag{2}$$

Here Φ_k denotes a generic k-fold excited determinant. For $k = 0$, the quantity is the correlation energy, not the scalar reference energy. For $k > 0$, it is the corresponding projected residual before setting it to zero. The subscript c restricts the expansion to connected contributions.

The Hamiltonian is generic: no Hartree–Fock zeros, additional orbital symmetries, spin integration, or finite orbital-space restrictions are imposed. All thirteen Hamiltonian vertices of Fig. 1 in Ref.[1] are retained. In particular, the bare one- and two-body excitation vertices are included in the singles and doubles equations. Each antisymmetrized topology is counted once, regardless of its algebraic sign, symmetry factor, or number of external-index permutations. This convention corresponds to the **non-HF antisymmetrized** column of Ref.[2], rather than to its separate Goldstone-diagram column. The count is not an operation count for an implementation using intermediates.

B. The Kállay–Surján integer string and tuple representation

Equation (8) and Sec. II A of Kállay and Surján (pp. 2947–2948) encode each diagram as a string of thirteen integers.[1] Grouping the first twelve integers into triplets gives the equivalent tuple-of-tuples form

$$D = ((\varepsilon_1, \iota_1, \pi_1), (\varepsilon_2, \iota_2, \pi_2), (\varepsilon_3, \iota_3, \pi_3), (\varepsilon_4, \iota_4, \pi_4), \eta). \tag{3}$$

The final integer $\eta \in \{0, 1, ..., 12\}$ selects the Hamiltonian vertex. Each active triplet describes one cluster vertex: ε_j is its excitation rank, ι_j is the number of lines contracted with the Hamiltonian, and π_j is the number of these that are particle lines. Thus the internal hole count is $\iota_j - \pi_j$, and the cluster's external particle and hole counts are $\varepsilon_j - \pi_j$ and $\varepsilon_j - \iota_j + \pi_j$, respectively. At most four cluster vertices occur; unused positions are padded on the right by $(0, 0, 0)$. Padding does not represent a rank-zero cluster operator.

Active triplets are lexicographically nondecreasing: excitation rank is ordered first, then internal-line count within equal ranks, then particle-line count within equal ranks and internal-line counts. Only the active triplets are ordered; the zero padding is appended afterwards. For example,

```
((2, 2, 1), (0, 0, 0), (0, 0, 0), (0, 0, 0), 5)
```

contains a single T_2 joined by one particle and one hole line to vertex $\eta = 5$. The latter also supplies one external particle and one external hole, so this is a doubles-equation diagram. In contrast,

```
((2, 4, 2), (0, 0, 0), (0, 0, 0), (0, 0, 0), 10)
```

has all four T_2 lines contracted with the deexcitation vertex and contributes to the correlation energy. Repeated identical **active** triplets are permitted and denote separate equivalent cluster vertices. The string records topology and line species, not orbital labels or the diagram weight.

C. A multiset representation

Represent a cluster vertex by (e, p, h), where e is its excitation rank and p and h are the numbers of particle and hole lines contracted with the Hamiltonian. An admissible connected vertex satisfies

$$1 \le e \le n, \quad 0 \le p, h \le e, \quad p + h \ge 1. \tag{4}$$

The equivalent descriptor in the Kállay–Surján representation is $(\varepsilon, \iota, \pi) = (e, p + h, p)$. Excitation operators do not contract with one another in this construction; each cluster vertex is attached directly to the Hamiltonian. Since the latter has at most four incident lines, at most four cluster vertices can occur.

Permuting cluster vertices does not produce a new canonical diagram. Sorting their descriptors gives a unique representative of a multiset, i.e., an unordered collection allowing repetitions. Conversely, a multiset satisfying the contraction constraints, together with its Hamiltonian vertex, determines the canonical descriptor. Thus there is a bijection between these multisets and the diagram objects counted here. The multiplicity of equivalent Wick contractions belongs to the diagram's weight, not to the number of canonical objects.

For a Hamiltonian vertex, write $(P, H; P_0, H_0)$ for its internal and external particle–hole line totals. The symbol H in this tuple is a hole count, not the Hamiltonian operator. Conservation of lines requires

$$\sum_j p_j = P, \quad \sum_j h_j = H, \quad P_0 + \sum_j (e_j - p_j) = H_0 + \sum_j (e_j - h_j) = k. \tag{5}$$

Consequently, the total cluster rank m is fixed by

$$m = \sum_j e_j = k + P - P_0 = k + H - H_0. \tag{6}$$

For the Hamiltonian in Eq. (1), $P, H \leq 2$ and $m \leq k + 2$. These small contraction bounds are what make a closed evaluation possible.

D. Explicit construction of the projected-equation diagrams

The following restates the algorithm of Sec. II A, p. 2948, of Ref.[1], using the notation above and making the connectedness and canonical-order tests explicit.

1. **Choose cluster ranks.** For each integer $m = \max(0, k-2), ..., k+2$, enumerate partitions $m = e_1 + ... + e_l$ with $0 \leq l \leq 4$ and $1 \leq e_1 \leq ... \leq e_l \leq n$. The empty partition has $m = 0$.
2. **Choose the Hamiltonian vertex.** For each η, retain the rank partition only when $P_0 + m - P = H_0 + m - H = k$ and the Hamiltonian has enough lower lines to attach all l clusters. The thirteen choices and their line totals are listed in Table I.
3. **Distribute internal lines.** Partition the $P + H$ lower Hamiltonian lines into l positive group sizes ι_j and permute those sizes over the chosen cluster ranks. Reject $\iota_j > 2e_j$.
4. **Assign particle and hole species.** Assign P particle markers (ones) and H hole markers (zeros) to those groups. Permutations inside an individual group are equivalent and are represented only once. The number of ones in group j is π_j. Retain only $0 \leq \pi_j \leq e_j$ and $0 \leq \iota_j - \pi_j \leq e_j$.
5. **Canonicalize and emit.** Retain lexicographically ordered active triplets, append zero padding and η, and output each canonical tuple once. With no clusters, the line constraints permit only the appropriate bare excitation vertex. There is no bare scalar reference-energy diagram.

The implementation in `generate_equation` uses an equivalent, less redundant version of the two line-distribution steps. For a fixed rank partition and Hamiltonian vertex it enumerates bounded compositions directly,

$$(p_1, ..., p_l) \in \left\{ \boldsymbol{p} : \sum_j p_j = P, 0 \leq p_j \leq e_j \right\}, \tag{7}$$

$$(h_1, ..., h_l) \in \left\{ \boldsymbol{h} : \sum_j h_j = H, 0 \le h_j \le e_j \right\}, \tag{8}$$

and forms $\big(e_j, p_j + h_j, p_j\big)$, rejecting $p_j + h_j = 0$. This avoids constructing permutations of individual binary line markers. Every valid diagram supplies one of the enumerated rank partitions and line assignments, so the construction is exhaustive. The hierarchical ordering eliminates permutations of indistinguishable cluster vertices, so the canonical output is unique. No sign or symmetry-factor calculation is needed merely to list the topologies.

III. Multiset generating functions

A. The multivariate construction

Let x mark the total cluster rank and let u and v mark internal particle and hole contractions. The ordinary generating function for the admissible cluster multisets is

$$\mathcal{M}_n(x, u, v) = \prod_{e=1}^{n} \prod_{\substack{0 \le p,h \le \min(e,2) \\ p+h \ge 1}} \big(1 - x^e u^p v^h\big)^{-1}. \tag{9}$$

Each factor is a geometric series whose exponent records how many times one vertex type occurs. This is the standard product construction for unlabelled multisets, as described in Sec. I.2 of Ref.[3]. Restricting p, h to two loses no diagram because the Hamiltonian cannot supply more contractions of either kind. Define

$$A^n_{PH}(x) = [u^P v^H] \mathcal{M}_n(x, u, v). \tag{10}$$

The notation $[x^j]$ denotes coefficient extraction, with coefficients of negative powers taken as zero for an ordinary polynomial. We have $A^n_{00} = 1$, and particle–hole reflection gives $A^n_{PH} = A^n_{HP}$. Only five nonempty contraction totals need to be evaluated: $(1, 0)$, $(2, 0)$, $(1, 1)$, $(2, 1)$, and $(2, 2)$.

B. Explicit contraction polynomials

Introduce the rank polynomials

$$E_n(x) = \sum_{e=1}^{n} x^e, \quad B_n(x) = \sum_{e=2}^{n} x^e, \tag{11}$$

where an empty sum is zero. A contraction type requiring rank at least one contributes E_n; a type requiring rank at least two contributes B_n. For two copies of the same contraction type with minimum rank one, the ranks form an unordered pair and contribute

$$Q_n(x) = \sum_{1 \le a \le b \le n} x^{a+b} = \frac{E_n(x)^2 + E_n(x^2)}{2}. \tag{12}$$

The product $E_n(x)^2$ counts ordered pairs. Pairs with different ranks occur twice, whereas equal-rank pairs occur once; the added diagonal term $E_n(x^2)$ therefore makes division by two exact. This is the two-element multiset construction of Sec. I.6.1 in Ref.[3], not a division by a diagram symmetry factor.

Suppressing the superscript n on A and writing $E = E_n$, $B = B_n$, and $Q = Q_n$, direct coefficient extraction from Eq. (9) gives

$$\begin{aligned} A_{10} &= E, \\ A_{20} &= B + Q, \\ A_{11} &= E + E^2, \\ A_{21} &= B + BE + E^2 + QE, \\ A_{22} &= B + 2EB + B^2 + Q + 2BQ + E^3 + Q^2. \end{aligned} \tag{13}$$

For example, the four terms of A_{21} correspond to the partitions of the contraction vector

$$(2,1), \quad (2,0)+(0,1), \quad (1,1)+(1,0), \quad (1,0)+(1,0)+(0,1). \tag{14}$$

In the last case the two $(1,0)$ vertices form an unordered pair, which explains the factor Q. Similarly, the nine vector partitions of $(2,2)$ give the seven grouped terms in A_{22}. The terms $2EB$ and $2BQ$ each combine a pair of particle–hole reflections; Q counts two $(1,1)$ vertices; and Q^2 counts two $(1,0)$ vertices and two $(0,1)$ vertices. This reduction avoids any sum over four-element excitation-rank partitions.

C. Summation over Hamiltonian vertices

Table I: Hamiltonian contraction totals and coefficient shifts. The labels η follow Fig. 1 of Kállay and Surján.[1] Reflected vertices remain distinct diagrams and are included by the factor two.

η	$(P, H; P_0, H_0)$	Polynomial	Extract
$0, 1$	$(1,0;1,0)$ and reflection	$2A_{10}$	$[x^k]$
2	$(1,1;0,0)$	A_{11}	$[x^{k+1}]$
$3, 4$	$(2,0;2,0)$ and reflection	$2A_{20}$	$[x^k]$
5	$(1,1;1,1)$	A_{11}	$[x^k]$
$6, 7$	$(2,1;1,0)$ and reflection	$2A_{21}$	$[x^{k+1}]$
$8, 9$	$(1,0;2,1)$ and reflection	$2A_{10}$	$[x^{k-1}]$
10	$(2,2;0,0)$	A_{22}	$[x^{k+2}]$
11	$(0,0;1,1)$	1	$[x^{k-1}]$
12	$(0,0;2,2)$	1	$[x^{k-2}]$

Using Eq. (6), the thirteen Hamiltonian vertices contribute as listed in Table I. Their sum is the rank-generating polynomial

$$\begin{aligned} G_n(x) = {} & 2A_{10} + 2A_{20} + A_{11} + x^{-1}A_{11} + 2x^{-1}A_{21} \\ & + 2xA_{10} + x^{-2}A_{22} + x + x^2, \end{aligned} \tag{15}$$

and the desired count is

$$N(n,k) = [x^k]G_n(x), \quad 0 \le k \le n. \tag{16}$$

Although Eq. (15) uses negative powers to express the coefficient shifts, the polynomials multiplying them start at sufficiently high degree that G_n has no negative powers. The terms $x + x^2$ are the bare excitation vertices. There is no bare constant term because E_{ref} is excluded. Together, Eq. (13) and Eq. (15) constitute a finite, non-enumerative coefficient formula for the canonical count.

IV. Complete equations and eventual quasipolynomiality

A. Rational generating function

Since every diagram in the rank-k equation has total cluster rank at most $k+2$, no individual cluster rank can exceed that bound. Thus the equation is complete once $n \geq k+2$, in agreement with Ref.[2]. Denote its count by $S(k)$. For a fixed coefficient, the finite rank polynomials may then be replaced by their formal power-series limits,

$$E(x) = \frac{x}{1-x}, \quad B(x) = \frac{x^2}{1-x}, \quad Q(x) = \frac{x^2}{(1-x)(1-x^2)}. \tag{17}$$

Substitution into Eq. (13) and Eq. (15) yields

$$\mathcal{S}(x) = \sum_{k\geq 0} S(k)x^k = \frac{3 + 9x + 5x^2 - 25x^3 - 8x^4 + 21x^5 - 5x^7 + x^8}{(1-x)^4(1+x)^2}. \tag{18}$$

The generating function is interpreted formally; no analytic convergence assumption is needed to extract its coefficients. Partial fractions give

$$\begin{aligned}\mathcal{S}(x) = x^2 - 3x - 5 - \frac{11}{4(1-x)} + \frac{103}{16(1-x)^2} \\ + \frac{11}{4(1-x)^3} + \frac{1}{4(1-x)^4} + \frac{5}{4(1+x)} + \frac{1}{16(1+x)^2}.\end{aligned} \tag{19}$$

The coefficient identity

$$[x^k](1-\alpha x)^{-r} = \alpha^k \binom{k+r-1}{r-1} \tag{20}$$

then evaluates each term directly. This is the rational-function method illustrated in Secs. 1.2 and 3.15 of Ref.[4].

B. Parity and the initial coefficients

A quasipolynomial is a function that agrees with one polynomial on each residue class modulo a fixed positive integer. Equivalently, it is a polynomial whose coefficients are periodic functions of the integer argument. A sequence is **eventually** quasipolynomial if this property holds after finitely many initial values. Proposition 4.4.1 of Ref.[5] characterizes quasipolynomials through proper rational generating functions whose poles are roots of unity. A polynomial part in an improper rational function changes only finitely many coefficients, giving the corresponding eventual statement.

In Eq. (18) the numerator is nonzero at both $x = 1$ and $x = -1$. The pole at 1 has order four and therefore contributes a polynomial of degree three. The pole at -1 has order two and contributes $(-1)^k$ times a polynomial of degree one. Hence the eventual quasipolynomial has degree three and minimal quasiperiod two. The parity effect has a concrete combinatorial origin: the equal-rank pairs in Eq. (12) introduce the factor $(1-x^2)^{-1}$.

Define

$$U(k) = \frac{2k^3 + 78k^2 + 529k + 321 + (3k+63)(-1)^k}{48}. \tag{21}$$

The complete-equation count, including every nonnegative rank, is

$$S(k) = U(k) - 5\delta_{k,0} - 3\delta_{k,1} + \delta_{k,2}, \tag{22}$$

where $\delta_{i,j}$ is the Kronecker delta. These finite-support corrections are exactly the coefficients of $-5 - 3x + x^2$ in Eq. (19). In particular, $S(0) = 3$, $S(1) = 15$, and $S(2) = 38$, whereas U alone would give 8, 18, and 37. For $k \geq 3$, the parity branches are

$$S(k) = \begin{cases} \dfrac{k^3 + 39k^2 + 266k + 192}{24} & k \text{ even} \\ \dfrac{k^3 + 39k^2 + 263k + 129}{24} & k \text{ odd.} \end{cases} \tag{23}$$

Thus the sequence is not a quasipolynomial at all nonnegative ranks without corrections. Its leading growth is $S(k) \sim \frac{k^3}{24}$, and the alternating contribution is only linear in k.

V. Finite-truncation boundary corrections

Only the cases $n = k$ and $n = k + 1$ differ from the complete equation. For $n = k + 1$, a removed cluster vertex must have rank $k + 2$. The total-rank bound forces it to be the sole cluster vertex. There is exactly one such diagram: the $(2, 2)$ contraction type in A_{22} attached to Hamiltonian vertex $\eta = 10$. Consequently $N(k + 1, k) = S(k) - 1$, including $k = 0$.

For $n = k \geq 1$, six additional diagrams contain a vertex of rank $k + 1$. Three have a single cluster vertex: one comes from $x^{-1}A_{11}$ and two from $2x^{-1}A_{21}$. The remaining three have total rank $k + 2$ and two cluster vertices, of ranks $k + 1$ and 1. They are the two reflected EB contributions and the unordered-pair contribution Q within $x^{-2}A_{22}$. These six objects are distinct, and no others can be removed because the total rank cannot exceed $k + 2$. Together with the rank-$(k + 2)$ diagram, they give a correction of seven.

For $n = k = 0$, there are no cluster vertices. All three complete-equation correlation-energy diagrams disappear, and the excluded scalar reference energy cannot supply a replacement. The complete result is therefore

$$N(n, k) = \begin{cases} 0 & n = k = 0 \\ S(k) - 7 & n = k \geq 1 \\ S(k) - 1 & n = k + 1 \\ S(k) & n \geq k + 2 \end{cases}, \quad 0 \leq k \leq n, \tag{24}$$

with S given by Eq. (22). The energy counts are 0, 2, and 3 at truncations $n = 0$, $n = 1$, and $n \geq 2$, respectively. The stable three topologies are the one-body contraction with T_1, the two-body contraction with T_2, and the two-body contraction with T_1^2.

VI. The full similarity-transformed Hamiltonian

A. Operator diagrams versus projected equations

For a pure excitation cluster operator, the connected-cluster identity gives

$$\overline{H}_{\mathrm{N}} = \exp(-T) H_{\mathrm{N}} \exp(T) = (H_{\mathrm{N}} \exp T)_{\mathrm{c}}. \tag{25}$$

This is Eq. (6) of Kállay and Surján, p. 2946.[1] The expression on the right is understood as a Wick expansion with only connected terms retained; it is not an instruction to contract every Hamiltonian annihilation line. The nested-commutator expansion terminates after four commutators for the present Hamiltonian, equivalently permitting at most four attached cluster vertices. Here we count the expanded, reference-normal-ordered operator before introducing intermediates, combining dressed tensors, or using solved CC equations to set residuals to zero.

The earlier equation count includes only diagrams that survive the action of $\overline{H}_{\mathrm{N}}$ on Φ_0, and then only the selected excitation rank. The full operator also contains external annihilation lines. They annihilate the reference but can act nontrivially on excited states. Thus summing $N(n,k)$ for $0 \le k \le n$ cannot count the full operator. Even summing all pure-excitation ranks would omit its annihilation sectors.

Here creation and annihilation refer to **quasiparticles relative to the reference**: particle creation creates an electron in a virtual orbital, whereas hole creation removes an electron from an occupied orbital. In the Hamiltonian tuple $(P, H; P_0, H_0)$, P, H count the available lower quasiparticle-annihilation lines and P_0, H_0 count the upper quasiparticle-creation lines. In the projected equations every lower line is internal; in the full operator an unused lower line remains external.

B. The same tuple determines every external line

Let an active cluster multiset have total rank m and contracted particle and hole totals s and t. The full-operator admissibility conditions are

$$s = \sum_j p_j \le P, \quad t = \sum_j h_j \le H, \quad p_j + h_j \ge 1. \tag{26}$$

The per-vertex rank bounds are unchanged. Replacing the two equalities in Eq. (5) by these inequalities is the essential generalization. The external signature, in the order created particles, created holes, annihilated particles, annihilated holes, is

$$\big(c_p, c_h, a_p, a_h\big) = (P_0 + m - s, H_0 + m - t, P - s, H - t). \tag{27}$$

All four numbers follow from the existing triplets and η; no extra tuple fields are required. Equivalent choices of which individual Hamiltonian lines of the same species are contracted affect the algebraic weight, not the unlabelled antisymmetrized topology. If explicit external orbital labels or permutation-resolved Wick contractions were wanted, additional information would be necessary, but those are different counting conventions. Kállay and Surján already note on p. 2948 that the integer strings can describe intermediates as well as final equation diagrams.

For example, the tuple

```
((2, 1, 1), (0, 0, 0), (0, 0, 0), (0, 0, 0), 10)
```

now represents a valid operator diagram. It has a single particle contraction from T_2 to the two-body deexcitation vertex and signature $(1, 2, 1, 2)$. It is a three-body operator contribution and vanishes when acting directly on the reference. It is excluded by the projected-equation algorithm, which would demand two particle and two hole contractions.

Electron-number conservation implies

$$g = c_p - a_p = c_h - a_h = m + P_0 - P = m + H_0 - H. \tag{28}$$

The integer g is the net excitation grade and need not be nonnegative. The usual operator body rank is $r = \frac{c_p+c_h+a_p+a_h}{2} = g + a_p + a_h$. The signature of a projected rank-k diagram is exactly $(k, k, 0, 0)$.

A direct enumeration of the full operator is therefore simple. For each Hamiltonian vertex, form all connected cluster types $(e, p+h, p)$ with $1 \le e \le n$, $p \le \min(e, P)$, and $h \le \min(e, H)$. Starting with the empty tuple and remaining capacities (P, H), recursively append nondecreasing cluster types that fit the remaining capacities. Subtract their particle and hole counts and emit every visited state, including the empty state, with the usual padding and Hamiltonian label. Each appended type consumes at least one lower line, so the recursion has depth at most four. No rank-k projection filter is applied. This is the procedure implemented by `generate_similarity_transformed`.

C. Sector-resolved generating function

The same multiset polynomials A^n_{st} enumerate these diagrams. Let x mark the net excitation grade and let y, z mark external particle and hole annihilation counts. The finite Laurent polynomial

$$\mathcal{B}_n(x, y, z) = \sum_{\eta=0}^{12} x^{P_0-P} \sum_{s=0}^{P} \sum_{t=0}^{H} y^{P-s} z^{H-t} A^n_{st}(x) \tag{29}$$

contains every operator topology once; the four Hamiltonian line totals in each summand are those of vertex η. The coefficient of $x^g y^{a_p} z^{a_h}$ counts the external signature $\big(g + a_p, g + a_h, a_p, a_h\big)$. Negative powers of x accommodate deexcitation sectors. The pure-creation part is precisely the earlier polynomial,

$$[y^0 z^0]\mathcal{B}_n(x, y, z) = G_n(x). \tag{30}$$

This identity also clarifies why G_n contains coefficients at equation ranks beyond the retained CC equations: it describes the entire action on the reference, not only the residuals with $k \le n$. For instance, for singles–doubles truncation,

$$G_2(x) = 3 + 14x + 31x^2 + 22x^3 + 13x^4 + 4x^5 + x^6. \tag{31}$$

Only its first three coefficients enter the CCSD energy and amplitude equations. The sum of all seven coefficients is 88, still smaller than the full operator count derived below.

D. Total count and annihilation sectors

Set all three markers to one in Eq. (29). Summing over the Hamiltonian capacities gives

$$C(n) = \mathcal{B}_n(1, 1, 1) = 13 + 16A_{10} + 6A_{20} + 5A_{11} + 4A_{21} + A_{22}, \tag{32}$$

where all A are now evaluated at $x = 1$. The constant 13 counts every bare Hamiltonian vertex, not a scalar reference energy. For $n \ge 1$,

$$E_n(1) = n, \quad B_n(1) = n - 1, \quad Q_n(1) = \frac{n(n+1)}{2}. \tag{33}$$

Substitution into Eq. (13) gives the closed result

$$C(n) = \begin{cases} 13 & n = 0 \\ \dfrac{n^4 + 18n^3 + 87n^2 + 106n + 12}{4} & n \ge 1. \end{cases} \tag{34}$$

Unlike the fixed-equation-rank count, this is an eventual **polynomial**, not a parity-dependent quasipolynomial. Summing over all allowed cluster ranks turns the unordered-pair factor into

$\frac{n(n+1)}{2}$. The leading $\frac{n^4}{4}$ term arises from $Q_n(1)^2$, describing two particle-only and two hole-only cluster vertices attached to $\eta = 10$. The exceptional value at zero follows from $B_0 = 0$, not $n - 1$. At zero truncation the operator is simply H_{N}, with all thirteen vertices. At positive truncation, the generated correlation-energy scalar is included, while the original scalar E_{ref} remains excluded.

More detail is obtained by summing only over g. Define

$$C_{a_p,a_h}(n) = [y^{a_p} z^{a_h}]\mathcal{B}_n(1, y, z). \tag{35}$$

The six sector classes under particle–hole reflection are listed in Table II. The two members of a reflected pair are distinct sectors with the same count, not a single combined entry. Their multiplicities therefore give

$$C(n) = C_{0,0}(n) + 2C_{1,0}(n) + 2C_{2,0}(n) + C_{1,1}(n) + 2C_{2,1}(n) + C_{2,2}(n). \tag{36}$$

Table II: Expanded operator counts resolved by external quasiparticle annihilation counts and summed over all net excitation grades. At $n = 2$, the total is $88 + 2(31) + 2(7) + 12 + 2(3) + 1 = 183$.

(a_p, a_h)	Count per sector, $n \geq 1$	$n = 2$
$(0, 0)$	$\frac{n^4+14n^3+47n^2+22n-8}{4}$	88
$(1, 0)$, $(0, 1)$	$\frac{n^3+8n^2+11n}{2}$	31
$(2, 0)$, $(0, 2)$	$\frac{n^2+5n}{2}$	7
$(1, 1)$	$(n + 1)(n + 2)$	12
$(2, 1)$, $(1, 2)$	$n + 1$	3
$(2, 2)$	1	1

For example, the $(1, 0)$ sector is $2 + 3E + A_{11} + A_{20} + A_{21}$, and the $(0, 0)$ sector is $2 + 4A_{10} + 2A_{20} + 2A_{11} + 2A_{21} + A_{22}$, all at $x = 1$. These expressions, together with Eq. (13), directly give the first two rows of Table II. The remaining sectors are $1 + E + A_{20}$, $2 + 2E + A_{11}$, $1 + E$, and 1, respectively. They also apply at $n = 0$ when the finite-polynomial values $E = B = Q = 0$ are used rather than the positive-n polynomial continuations.

For reference, the totals at truncations $n = 0, 1, 2, 3, 4, 5$ are $13, 56, 183, 420, 809, 1398$. These are counts of expanded topologies across all operator body ranks. At $n \geq 1$ the maximum body rank is $4n - 2$: a bare b-body vertex with c contractions has body rank $b + m - c$; connectedness gives $m \leq cn$, and $c \leq 2b \leq 4$. The bound is attained by four T_n vertices singly attached to the two-body deexcitation vertex. It is consistent with the fact that a singles-only similarity transformation does not increase the two-body rank. Counting only a specified body-rank block, an EOM matrix element, or a factorized intermediate representation would be a different enumeration.

VII. Extension to extended coupled cluster

A. Two successive connectivity conditions

Use a single, common truncation rank n for the excitation and deexcitation operators, retaining all amplitudes of ranks $1, ..., n$:

$$T = \sum_{e=1}^{n} T_e, \quad S = \sum_{r=1}^{n} S_r, \quad \mathcal{K} = \exp(S)\exp(-T)H_{\mathrm{N}}\exp(T)\exp(-S). \tag{37}$$

Each S_r contains r particle and r hole quasiparticle annihilators. The common cutoff does not identify the two sets of amplitudes: S and T are independent, and no adjoint relation is assumed. Normal order the inner transform first. Since a pure annihilation operator S placed on the right of a normal-ordered term cannot contract with it, the final $\exp(-S)$ cancels the outer S vertices that have no contraction with that term. Thus

$$\mathcal{K} = \Big[\exp(S)(H_{\mathrm{N}}\exp(T))_{C_T}\Big]_{C_S}, \tag{38}$$

where C_T requires every T to touch H_{N} directly, and C_S requires every S to touch at least one creation line of the resulting Hamiltonian–T subgraph. This also follows by applying the connected commutator argument separately to the two similarity transformations. Permitted contractions are H–T, H–S, and T–S; there are no T–T or S–S contractions. An H–S contraction uses an **upper creation line** of H, whereas an H–T contraction uses a **lower annihilation line**. Every T still needs its own direct connection to H: mere connectedness of the final graph would incorrectly admit T vertices linked to H only through an S vertex.

B. Original and canonical amplitude coordinates

The original, or noncanonical, ECC coordinates are the independent amplitudes (T, S). They parameterize a right state $\exp(T)\Phi_0$ and a left state represented by the bra $\langle\ \Phi_0\ |\exp(S)\exp(-T)$. Their overlap is one, since

$$\langle\Phi_0|\exp(S)\exp(-T)\exp(T)|\Phi_0\rangle = 1. \tag{39}$$

This is a bivariational parameterization, not a requirement that the bra be the Hermitian adjoint of the ket. Its energy is

$$E_{\mathrm{NC}}(T, S) = \langle\Phi_0|\exp(S)\exp(-T)H_{\mathrm{N}}\exp(T)|\Phi_0\rangle. \tag{40}$$

Here "NC" means noncanonical ECC, not ordinary CC with a linear left parameterization $1 + \Lambda$: the full exponential $\exp(S)$ is retained. In particular, the literal double transform of Eq. (37) uses these original coordinates. Its two successive connectivity conditions must be distinguished from the stronger doubly-linked energy theorem below.

Define a new pure excitation operator Σ, leaving the deexcitation operator S unchanged, by

$$\Sigma\Phi_0 = Q\exp(S)T\Phi_0, \quad T\Phi_0 = Q\exp(-S)\Sigma\Phi_0, \quad Q = 1 - |\ \Phi_0\ \rangle\langle\ \Phi_0\ |. \tag{41}$$

The projector removes the reference component; Σ and T have no scalar amplitudes. These maps are inverses because $\exp(S)\Phi_0 = \exp(-S)\Phi_0 = \Phi_0$. They are linear in the excitation amplitudes at fixed S, but nonlinear as transformations of the joint amplitude coordinates. This is the map in Sec. III A, Eqs. (11)–(14), p. 5 of arXiv:2003.06796,[6] with Σ denoting that reference's T' and with the deexcitation operator written directly as S.

The canonical-coordinate energy is the same scalar functional expressed in the new variables:

$$E_{\mathrm{C}}(\Sigma, S) = E_{\mathrm{NC}}(T(\Sigma, S), S). \tag{42}$$

Its diagrammatic simplification is

$$E_{\mathrm{C}}(\Sigma, S) = \langle\Phi_0|\exp(S)\exp(-\Sigma)H_{\mathrm{N}}\exp(\Sigma)|\Phi_0\rangle_{\mathrm{DL}}, \tag{43}$$

where every S vertex must touch H or at least **two distinct** Σ vertices. Terms in which an S is attached only to a single excitation vertex are absorbed into the coordinate map. The negative terms in the inverse map cancel those single-source attachments in the energy expansion. They are not neglected contributions to an approximate energy. Consequently, Eq. (43) is generally **not** $E_{\mathrm{NC}}(\Sigma, S)$ evaluated with the same numerical amplitudes. It equals the original energy at the **mapped** amplitudes $T(\Sigma, S), S$. Nor does the energy theorem license filtering the full literal operator $\mathcal{K}(\Sigma, S)$ in the same way: an operator expressed in canonical coordinates is first obtained by substituting $T(\Sigma, S)$.

1. Why these coordinates are called canonical

The word refers to the Hamiltonian structure of the time-dependent bivariational equations, not to canonical Hartree–Fock orbitals or to sorting integer tuples. In dual excitation/deexcitation bases, write the amplitudes of Σ, S as σ_μ, s_μ. The kinetic one-form of the bivariational action obeys

$$\theta = \langle \Phi_0 | \exp(S) dT | \Phi_0 \rangle = df - \sum_\mu \sigma_\mu ds_\mu, \quad f = \langle \Phi_0 | \exp(S) T | \Phi_0 \rangle. \tag{44}$$

This follows by differentiating f, using commutativity of the pure deexcitation operators and Eq. (41). Up to an exact differential, it is the canonical one-form $\sum_\mu s_\mu d\sigma_\mu$. With $\hbar = 1$, the corresponding equations have the canonical form

$$i\frac{d\sigma_\mu}{dt} = \frac{\partial E_{\mathrm{C}}}{\partial s_\mu}, \quad -i\frac{ds_\mu}{dt} = \frac{\partial E_{\mathrm{C}}}{\partial \sigma_\mu}. \tag{45}$$

The original amplitudes T, S generally do not give this simple canonical pairing. Ordinary graph canonicalization, used below to choose a unique representative, is unrelated to this symplectic meaning of "canonical."

2. Equivalence at a common rank-complete truncation

A deexcitation acting on a rank-e excited state can only lower its excitation rank. Therefore both maps in Eq. (41) preserve the space containing **all** excitation amplitudes of ranks $1, ..., n$. The map is triangular in rank with identity diagonal: $\Sigma_n = T_n$, and only higher-rank T amplitudes can contribute to $\Sigma_k - T_k$. The unchanged S retains the same cutoff. Thus the original and canonical formulations at this common, rank-complete truncation are two invertibly related coordinate descriptions of the **same** truncated ECC model, not two different correlation approximations. The corresponding result is Theorem 5 in Sec. III D, p. 7 of the same arXiv version.[6] Their stationary solutions map into one another and have the same energy. Algebraically, the stationary equations are related by an invertible Jacobian, although their written diagram expansions differ.

For ECC singles–doubles, the entire coordinate transformation reduces to

$$\Sigma_2 = T_2, \quad \Sigma_1 = T_1 + (S_1 T_2)_{\mathrm{exc},1}, \tag{46}$$

$$T_2 = \Sigma_2, \quad T_1 = \Sigma_1 - (S_1 \Sigma_2)_{\mathrm{exc},1}. \tag{47}$$

Here $(S_1 T_2)_{\mathrm{exc},1}$ is the pure rank-one excitation operator whose action on Φ_0 is the rank-one part of $S_1 T_2 \Phi_0$; the S_1 annihilators are fully contracted. Contributions from $S_2 T_2$ or $S_1^2 T_2$ are scalars and are removed by Q. This makes the reduction of energy diagrams transparent: single-source S_1 –T_2 attachments are incorporated into Σ_1. An energy S_2 cannot attach only to one T_e with $e \leq 2$, since that T_e must already use at least one line to touch H. More generally, a fully contracted S_r

attached only to one T_e requires $e > r$. At singles-only truncation there are no such attachments, the coordinate map is the identity, and the two energy counts agree.

Rank completeness matters. Additional selected-excitation or orbital-space restrictions need not be closed under the rank-lowering coordinate map; then equivalence must be checked separately. Equality of just the two maximum ranks is not by itself a proof for such restricted models. In the generic rank-complete convention used here, the reduction from 1503 to 907 ECCSD energy topologies reflects a change of coordinates, not a change in the stationary energy. The construction below uses T to name its excitation-vertex slots; for the canonical-coordinate energy those slots are interpreted as Σ vertices.

C. An incidence extension of the integer string

Retain an ordinary canonical operator tuple D_T, with $l \leq 4$ active T vertices, as the Hamiltonian–T subgraph. Its triplets remain $\big(e_j, p_j + h_j, p_j\big)$, but these internal-line counts now mean specifically ***H–T* contractions**, not all internal lines incident on $T^{(j)}$ in the final ECC graph. The available creation-line capacities at sources $j = 0, 1, ..., l$ are

$$\big(\gamma_0^p, \gamma_0^h\big) = (P_0, H_0), \quad \big(\gamma_j^p, \gamma_j^h\big) = \big(e_j - p_j, e_j - h_j\big) \ \text{ for } \ j \geq 1. \tag{48}$$

Source zero is H, and sources $1, ..., l$ are the active T vertices in their canonical order. Describe S vertex a by its rank r_a and one pair of edge counts per source,

$$\omega_a = (r_a, (b_{a0}, b_{a1}, ..., b_{al})), \quad b_{aj} = \big(u_{aj}, v_{aj}\big). \tag{49}$$

The entries count particle and hole contractions from source j to this S vertex. The full representation is the nested tuple

$$D_{\mathrm{ECC}} = \big(D_T, (\omega_1, ..., \omega_q)\big). \tag{50}$$

No individual orbital labels are introduced. The original zero padding of D_T is retained, but the S tuple is variable length and has no padding. The necessary and sufficient incidence constraints are

$$1 \leq r_a \leq n, \quad u_{aj}, v_{aj} \in \mathbb{Z}_{\geq 0}, \tag{51}$$

$$\sum_j u_{aj} \leq r_a, \quad \sum_j v_{aj} \leq r_a, \quad \sum_j \big(u_{aj} + v_{aj}\big) \geq 1, \tag{52}$$

$$\sum_a u_{aj} \leq \gamma_j^p, \quad \sum_a v_{aj} \leq \gamma_j^h. \tag{53}$$

The first two row inequalities are the rank bounds; the last enforces the outer connectedness condition. The column inequalities prohibit using a source line twice. Along with the validity of D_T, these conditions construct an admissible species-coloured contraction multigraph. All remaining lines are external. Selecting individual indistinguishable lines within a species changes Wick multiplicities and symmetry factors, not the topology counted here.

Adding only an S triplet (r_a, ι_a, π_a) would lose essential information: its entries give row totals but not the destinations of the edges. For a concrete counterexample, take

```
D_T = ((2, 2, 1), (2, 2, 1), (0, 0, 0), (0, 0, 0), 10)
```

and two S_1 vertices. Both T_2 vertices have one remaining particle and one remaining hole. Two inequivalent incidence matrices, with columns $H, T^{(1)}, T^{(2)}$ labelling the two rank-two vertices, are

$$B_A = \begin{pmatrix} (0,0) & (1,1) & (0,0) \\ (0,0) & (0,0) & (1,1) \end{pmatrix}, \quad B_B = \begin{pmatrix} (0,0) & (1,0) & (0,1) \\ (0,0) & (0,1) & (1,0) \end{pmatrix}. \tag{54}$$

In B_A, each S_1 closes the remaining pair on a single T_2. In B_B, each S_1 connects the particle from one T_2 to the hole from the other. Both have the same original T triplets, the same S triplets $(1,2,1)$, and even the same particle–hole degree at every vertex. Nevertheless, they are different scalar topologies. Both belong to the literal ECC energy; only B_B satisfies the stronger canonical-coordinate doubly-linked condition.

D. External signature, termination, and energy projection

Write the inner operator signature as (C_p, C_h, A_p, A_h) and define

$$R_S = \sum_a r_a, \quad U = \sum_{a,j} u_{aj}, \quad V = \sum_{a,j} v_{aj}. \tag{55}$$

The final signature is

$$(c_p, c_h, a_p, a_h) = (C_p - U, C_h - V, A_p + R_S - U, A_h + R_S - V). \tag{56}$$

In particular, its excitation grade is $g = g_T - R_S$, where $g_T = C_p - A_p = C_h - A_h$. This recovers a rank-selection rule analogous to Eq. (6): for a fixed target signature the total S rank is fixed, $R_S = g_T - g$. Its required particle and hole contraction totals are $U = C_p - c_p$ and $V = C_h - c_h$. Negative values rule out that subgraph. The two annihilation-count equations in Eq. (56) must also hold.

There is a finite construction at a fixed common cutoff n, but there is **no four-vertex bound on S**. Each S consumes at least one available creation line, so, for $n \geq 1$,

$$q \leq C_p + C_h \leq 8n - 4, \quad R_S \leq n(C_p + C_h). \tag{57}$$

To see the creation-line bound, a bare b-body vertex with L lower lines, c contractions to T, and total attached rank m has $C_p + C_h = 2b - L + 2m - c$. Use $m \leq cn$, $c \leq L \leq 2b$, and $b \leq 2$. At $n = 2$, twelve singly attached S_1 vertices are possible in the full operator. Thus a common **rank** cutoff does not truncate either exponential at order n or impose four padded slots on both cluster families. At $n = 0$ there are no cluster vertices and $\mathcal{K} = H_{\mathrm{N}}$.

The ECC energy is the scalar projection

$$E_{\mathrm{ECC}} = \langle \Phi_0 | \mathcal{K} | \Phi_0 \rangle = \langle \Phi_0 | \exp(S) \exp(-T) H_{\mathrm{N}} \exp(T) | \Phi_0 \rangle. \tag{58}$$

Since $R_S - U \geq 0$ and $R_S - V \geq 0$, a scalar requires $A_p = A_h = 0$, $C_p = C_h = R_S$, and every S row fully contracted,

$$\sum_j u_{aj} = \sum_j v_{aj} = r_a, \quad \sum_a u_{aj} = \gamma_j^p, \quad \sum_a v_{aj} = \gamma_j^h. \tag{59}$$

These equalities greatly reduce the energy enumeration. With $R_T = \sum_j e_j$, the common-truncation energy problem can be organized by pairs of rank partitions satisfying

$$1 \leq e_j, r_a \leq n, \quad R_T - R_S = P - P_0 = H - H_0, \quad |R_T - R_S| \leq 2. \tag{60}$$

In addition, $R_S = C_p = C_h \leq 4n - 2$ for $n \geq 1$, so the energy has at most $4n - 2$ deexcitation vertices. Sharing a cutoff does **not** require $R_T = R_S$: the Hamiltonian supplies their grade difference. It also does not allow us to discard inner pure-excitation ranks above n, since they can be closed by

several S vertices. The incidence data and residual column-permutation quotient remain necessary. For the canonical-coordinate energy only, add the row test

$$u_{a0} + v_{a0} > 0 \quad \text{or} \quad \left|\left\{j \geq 1 : u_{aj} + v_{aj} > 0\right\}\right| \geq 2. \tag{61}$$

Two lines attached to the **same** T do not meet the second alternative. No statement here identifies this filtered energy with the full operator in untransformed T, S coordinates.

E. Unique representatives and an explicit construction algorithm

For a fixed labelled D_T, sort the S rows lexicographically. This quotients permutations of equivalent deexcitation vertices, but is not yet sufficient: two identical T triplets can exchange their entire incidence columns. Let

$$\Gamma_{D_T} = \left\{\rho \in \mathfrak{S}_l : \left(e_{\rho(j)}, p_{\rho(j)}, h_{\rho(j)}\right) = \left(e_j, p_j, h_j\right) \ \text{ for all } \ j\right\}. \tag{62}$$

Equivalently, Γ_{D_T} is the direct product of permutation groups on the multiplicity classes of equal T triplets. The four slots give the crude order bound $4! = 24$, but the present line capacities sharpen it to $\left|\Gamma_{D_T}\right| \leq 4$. Every connected type consumes at least one particle or hole line, so no type can occur more than twice when $P, H \leq 2$. At most two multiplicity-two classes occur, giving $2! \cdot 2! = 4$. The bound is attained by two particle-only and two hole-only triplets attached to vertex $\eta = 10$. Keep the Hamiltonian column fixed. For every ρ in this group, permute the T columns simultaneously in all S rows, sort the rows, and select the lexicographically smallest row tuple. Together with D_T this is the unique ECC representative. Permutations of unlike T triplets have already been removed by canonicalizing the inner subgraph; particle and hole colours and the T and S vertex families are never interchanged.

The resulting extension of the Kállay–Surján construction is:

1. Generate each canonical inner operator tuple D_T using the partial-contraction algorithm of the preceding section, not just the retained projected CC equations. Compute its source capacities.
2. Form every candidate S row of rank $1, ..., n$ satisfying the row bounds and the individual source capacities. For energy diagrams, impose full row contraction at this stage and discard inner annihilation sectors.
3. Starting from the empty row tuple, recursively append nondecreasing candidate rows that fit the remaining source capacities. Subtract their edge counts column by column. Emit every visited state for the full operator; emit only states exhausting the capacities for the energy. Reusing a row is permitted and represents another identical S vertex.
4. Canonicalize each emitted state under Γ_{D_T}, sorting its rows after each column permutation, and retain each representative once.
5. Apply any requested external-signature filter using Eq. (56). For a preselected signature one can instead enumerate nondecreasing S ranks summing to $g_T - g$ and distribute the required U, V by bounded incidence matrices, directly paralleling the rank-partition and line-distribution steps of the original algorithm. Apply Eq. (61) only when the canonical-coordinate energy is requested.

This construction is exhaustive because deleting all S vertices from any valid diagram recovers one generated inner subgraph, and its remaining edges determine an allowed multiset of rows. It is unique because any isomorphism between two such diagrams restricts to an automorphism

of that fixed inner subgraph, namely an element of Γ_{D_T}, together with a permutation of S rows. Conversely, each of those permutations preserves the represented topology. Testing at most four admissible column permutations therefore suffices; a factorial search over all S vertex permutations is unnecessary. The number of distinct output diagrams can nevertheless be very large.

If the desired objects are ECC **stationarity equations** rather than operator sectors, start with the appropriate scalar energy graphs and mark a vertex of the amplitude rank being differentiated. Canonicalize with that vertex distinguished; removing it exposes the residual indices. Equivalent choices of the marked vertex supply derivative multiplicities, not additional unique residual topologies. This applies separately to variations of S and T (or S, Σ in canonical coordinates). Simply filtering annihilation and creation sectors of the literal transformed operator should not be identified with both families of ECC stationarity equations without the corresponding variation argument.

F. Counting without listing graph representatives

There is a useful independent check on the orbit enumeration. For a fixed inner tuple, let $\mathcal{R}$ be the finite set of allowed S row types and associate with a row ω the monomial

$$w(\omega) = z^{r_\omega} \prod_{j=0}^{l} x_j^{u_{\omega j}} y_j^{v_{\omega j}}. \tag{63}$$

An ordinary multiset product $\prod_{\omega \in \mathcal{R}} (1 - w(\omega))^{-1}$ counts rows with **labelled source columns**. It overcounts whenever Γ_{D_T} has nontrivial orbits; dividing that count by the group order is also incorrect because stabilizers vary. For $\rho \in \Gamma_{D_T}$, a fixed multiset must contain every row in a ρ-orbit with equal multiplicity. Its generating function is therefore

$$F_\rho = \prod_{O \in \mathcal{R} \,/\, \langle \rho \rangle} \left(1 - \prod_{\omega \in O} w(\omega) \right)^{-1}. \tag{64}$$

Here the product runs over row-type orbits under the cyclic subgroup generated by ρ, not over its cycles on individual source columns. Truncate source degrees to γ_j^p, γ_j^h. For each retained term, z records R_S, while the sums of the x and y degrees give U, V. Classifying by Eq. (56) gives fixed-diagram counts in each external sector. Burnside's lemma then gives the number of inequivalent diagrams for that sector as the average over ρ. Source capacities are constant on equivalent T columns, so this truncation and sector selection are invariant under the group. Summing the averages over all inner tuples counts the complete ECC operator. Energy counts follow by using fully contracted row types and retaining exactly the source capacities; the stronger energy condition can also be imposed directly on $\mathcal{R}$.

This is a finite coefficient algorithm, not a quartic closed form analogous to Eq. (34). Its extra orbit structure is precisely the information lost by an attempted extension using only cluster triplets and total line counts.

G. Implementation and small-truncation checks

The unified `generate_kallay.py` implements the CC, full single-transform, and ECC constructions, including the incidence tuples and explicit algorithm. `generate_ecc(n)` returns the literal full operator

with a common cutoff for both cluster families; `ecc_external_signature` returns its four external line counts. `generate_ecc_energy(n)` closes all lines, and its optional `canonical_coordinates=True` selects the stronger energy theorem, with the amplitude interpretation described above. The default equation CLI and CC tuple format are unchanged; additional `--mode` choices select `hbar`, `ecc`, or `ecc-energy`. The canonical-coordinate option is restricted to the last mode. JSON arrays from these modes can be read directly by Typst.

The accompanying `lib.typ` now renders partially contracted operators and nested ECC tuples as well as equation diagrams. The extended layout uses explicit creation/annihilation ports, preserves every incidence entry, and leaves unused ports as short external stubs. T vertices lie below H and S vertices above it; H is displaced sideways when necessary to keep T–S edges away from its vertices. The original equation layout remains the default for undecorated, fully contracted Hamiltonian annihilation sectors. Rendering applies neither an energy-connectivity filter nor a coordinate transformation. Tests reconstruct the incidence matrices and signatures from the actual rendering segments for all 183 rank-two single-transform tuples, all 395 singles-only ECC operator tuples, and all 1503 original-coordinate ECCSD energy tuples. Additional checks exercise malformed inputs and a twelve-S-vertex diagram.

Table III: ECC topology counts at a common rank-complete truncation n for excitation and deexcitation amplitudes. "Canonical" refers to amplitude coordinates, not to graph labelling. All bare H_{N} vertices are retained, and the reference scalar is excluded. The starred full-operator count is obtained by Burnside coefficient counting only, without listing its millions of representatives. All other entries were also explicitly enumerated.

n	Full operator	Energy, original	Energy, canonical
0	13	0	0
1	395	20	20
2	8,743,171*	1,503	907

The test file `tests/test_generate_ecc.py` independently implements Eq. (64) and the Burnside average. Full-operator counts resolved by both inner tuple and external signature agree with explicit enumeration at common cutoffs $n = 0, 1$. Both energy variants agree with the orbit-product counts at $n = 2$. Additional checks cover row and column relabellings and the incidence examples. Unequal-cutoff helper functions are retained privately for diagnostic checks only: switching off either cluster family recovers a single similarity transformation, and an additional asymmetric case checks larger graphs against explicit enumeration. These are not additional members of the common-truncation ECC hierarchy presented here. The full $n = 2$ count illustrates why the explicit generator is intended as a small-truncation reference implementation rather than a scalable way to store all ECC operators. None of these checks evaluates signs, symmetry factors, or numerical ECC energies.

VIII. Published counts and numerical verification

A. Relation to Musiał, Kucharski, and Bartlett

The variable n in Table 1 of Ref.[2] denotes the equation rank, not the CC truncation. In the notation used here, their three models therefore correspond to $n = k$, $n = k + 1$, and $n = k + 2$. Their auxiliary polynomial for antisymmetrized diagrams is

$$N_a(k) = \frac{k^3}{24} + 13\frac{k^2}{8} + 133\frac{k}{12} + 1. \tag{65}$$

For $k \geq 3$, our formulas imply

$$S(k) = \begin{cases} N_a(k) + 7 & k \text{ even} \\ N_a(k) + \frac{9}{2} - \frac{k+1}{8} & k \text{ odd.} \end{cases} \tag{66}$$

Together with Eq. (24), this gives Table IV. The base-model formula and the six-then-one increase on adding higher clusters coincide with the counts and discussion of Ref.[2].

Table IV: Non-HF antisymmetrized counts for $k \geq 3$, expressed using the auxiliary polynomial of Musiał et al.[2] The two positive half-integer terms in the higher-truncation odd-rank rows correct apparent sign errors in the printed table; see the text.

Truncation	k even	k odd	Unified form
$n = k$	N_a	$N_a - \frac{5}{2} - \frac{k+1}{8}$	$S(k) - 7$
$n = k + 1$	$N_a + 6$	$N_a + \frac{7}{2} - \frac{k+1}{8}$	$S(k) - 1$
$n \geq k + 2$	$N_a + 7$	$N_a + \frac{9}{2} - \frac{k+1}{8}$	$S(k)$

A literal comparison requires noting two apparent typographical errors. The printed non-HF antisymmetrized odd-rank entries for truncations $k + 1$ and $k + 2$ contain $-\frac{7}{2}$ and $-\frac{9}{2}$, respectively, in place of $+\frac{7}{2}$ and $+\frac{9}{2}$. The printed signs would make the count decrease by one upon each enlargement of the cluster space, contrary both to the inclusion of diagram sets and to the increments of six and one stated in Sec. 2 of that paper. For example, at $k = 3$ the printed entries give 46 and 45, while direct enumeration and Eq. (24) give 53 and 54, following the base count 47. The corrected signs in Table IV reconcile these statements. No claim about corrections to the separate HF or Goldstone columns is needed here.

There is also a genuine convention difference at low rank. Table 1 of Ref.[2] explicitly omits the constant term in the doubles equation. This term is the bare excitation integral, not the scalar reference energy. Because it is retained here, our doubles counts $(31, 37, 38)$ at truncations $(2, 3, 4)$ exceed their counts $(30, 36, 37)$ by one. Their Table 1 is restricted to equation ranks above one. In Table 4, the non-HF CCS singles count similarly omits its bare one-body excitation term and is seven rather than eight. These differences are independent of the exclusion of E_{ref} in Eq. (1).

B. Reproducible enumeration checks

The closed form and the finite generating polynomials were checked against `generate_equation(n, k)` in the accompanying Python implementation of the Kállay–Surján construction. Every pair $0 \leq k \leq n \leq 20$ was tested, giving 231 distinct truncation–equation comparisons. The checks also

compare the contribution of each of the thirteen Hamiltonian vertices, rather than only their sum. In addition, coefficients through degree 80 obtained from the multiset polynomials were checked against both Eq. (18) and Eq. (19). All comparisons use exact integer or rational arithmetic. The accompanying regression test `tests/test_diagram_count_derivation.py` records these checks; no numerical fit is used in the derivation. The positive-rank values also agree with Table I of Kállay and Surján, p. 2948, which uses the same bare-vertex convention as the present generator.[1]

The full-operator implementation was independently checked for every $n = 0, ..., 20$ by directly multiplying the multivariate geometric factors in Eq. (9), retaining particle and hole degrees through two. Applying Eq. (29) to those coefficients gives the expected count for each Hamiltonian label and complete external signature. These counts agree with explicit tuple enumeration, Eq. (34), and Table II. Filtering the operator tuples to signature $(k, k, 0, 0)$ recovers the original equation generator's exact tuples for all 231 pairs, not merely their counts. `tests/test_similarity_transformed.py` records these comparisons, the maximum-body-rank check, and the explicit tuple examples. The formula can be evaluated without enumeration using `count_similarity_transformed(n)`; `external_signature` returns the four line counts of a valid tuple.

Table V: Canonical connected antisymmetrized counts $N(n, k)$. Bare singles and doubles vertices are included, and the scalar reference energy is excluded. Dashes denote ranks outside $0 \leq k \leq n$.

n	$k=0$	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
0	0	—	—	—	—	—	—
1	2	8	—	—	—	—	—
2	3	14	31	—	—	—	—
3	3	15	37	47	—	—	—
4	3	15	38	53	74	—	—
5	3	15	38	54	80	99	—
6	3	15	38	54	81	105	135

Representative values are given in Table V. Evaluating Eq. (22) and Eq. (24) requires a fixed number of arithmetic operations and no diagram enumeration. This is a constant arithmetic-operation count, not a claim of constant bit complexity for arbitrarily large integer inputs. Agreement of total counts is a useful regression check, but it does not by itself validate the algebraic weights of the diagrams.

IX. Conclusions

The canonical antisymmetrized diagrams of a one- and two-body CC Hamiltonian can be counted as multisets of cluster vertices subject to small particle–hole contraction totals. This representation reduces the problem to five explicit polynomials and a finite sum over Hamiltonian vertices. Their complete-equation generating function explains the known cubic, parity-dependent count through poles at 1 and -1. Its polynomial part supplies the low-rank corrections, while the maximum total cluster rank gives the finite-truncation corrections directly.

The result provides an alternative combinatorial derivation of the non-HF antisymmetrized formulas of Musiał et al., rather than a new replacement for those formulas. It also identifies

precisely which bare-vertex conventions must be reconciled when using published counts to validate an automatic diagram generator.

The same representation also counts the full similarity-transformed Hamiltonian. Allowing uncontracted Hamiltonian annihilation lines leaves the tuple format unchanged because the remaining external lines are inferable from it. A sector-resolved generating function then gives the quartic total count and recovers all projected equations as its annihilation-free part. Thus cubic parity dependence at fixed equation rank and quartic polynomial growth of the full operator have a common multiset origin, but concern different counting problems and different rank variables. For a second similarity transformation by a pure deexcitation operator, the required additional information is the particle–hole incidence of each new cluster with the existing subgraph. Multisets of these incidence rows, quotiented by the residual permutations of at most four excitation vertices, provide a unique representation and an exhaustive construction. A separate Burnside coefficient calculation validates the small-truncation counts. The distinction between the literal double transform and the stronger canonical-coordinate ECC energy theorem is essential: imposing the latter restriction indiscriminately would discard valid operator topologies. No simple polynomial total for general ECC truncations is claimed here.

Extending the Hamiltonian beyond two-body rank would require more cluster slots and additional contraction polynomials and larger permutation groups. The generating-function and incidence constructions remain starting points, but the degree, period, and group-size bounds obtained here rely on the present Hamiltonian and the specified pure excitation/deexcitation operators.

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Author declarations

Conflict of interest

[Author action: supply the authors' conflict-of-interest declaration.]

Author contributions

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Data availability

The integer counts reported here can be reproduced from the equations in this article and the Python generator and regression tests accompanying the manuscript source. [Author action: deposit these files in a persistent public archive and add its identifier before submission.]

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